

Coloratura: A Real-Time Audio-Reactive LED Visualization System

Abstract

Coloratura is a real-time audio-reactive lighting system designed to convert live sound input into synchronized RGB LED animations. The system employs a hybrid architecture consisting of a Raspberry Pi 5 for digital signal processing (DSP) and an Arduino microcontroller for precise LED timing control. By extracting amplitude, spectral, and rhythmic features from incoming audio and mapping them to dynamic lighting patterns, Coloratura offers a low-cost, open-source alternative to commercial audio-visualization platforms. The project aims to achieve end-to-end latency under 50 ms while maintaining visual smoothness and fidelity. This proposal outlines the system motivation, related technical background, design methodology, and development plan.

I. Introduction

Audio-reactive lighting systems are increasingly used in performance venues, interactive installations, and home entertainment. Existing commercial solutions—such as DMX lighting controllers, Philips Hue Sync, and PC-based visualization engines—provide high-quality effects but are often expensive, proprietary, or reliant on closed ecosystems. As a result, these systems are inaccessible to hobbyists, students, and researchers who require open, customizable platforms.

Coloratura addresses this gap by developing a real-time audio-reactive lighting system built with affordable, consumer-grade hardware. The Raspberry Pi 5 performs complex DSP operations

such as Fast Fourier Transform (FFT) analysis, amplitude tracking, and transient detection, while an Arduino Uno or Arduino Due handles deterministic LED control. This division of labor leverages the computational strengths of the Pi and the timing precision of the Arduino to achieve low latency and visually coherent lighting effects.

The primary objective of this project is to create a robust, extensible, and open-source framework for sound-responsive lighting that rivals the responsiveness and fidelity of commercial products.

II. Background

A. Hardware Platforms

The Raspberry Pi 5 includes a quad-core ARM Cortex-A76 processor at 2.4 GHz and supports real-time DSP workloads with improved I/O bandwidth and GPU performance [1], [2]. Audio input is captured using a USB microphone due to the Pi's lack of built-in analog-to-digital conversion [3].

The Arduino Uno complements the Pi by handling precise LED timing via libraries such as FastLED and Adafruit NeoPixel. Addressable LEDs like WS2812B or APA102 require strict microsecond-level timing, making microcontrollers ideal for these tasks [4]–[6].

B. Audio Capture and Latency

Low latency is critical for perceptual alignment between audio and visual responses. The Raspberry Pi supports ALSA, JACK, and PortAudio, which can achieve latencies between

10–50 ms depending on buffer configuration [7], [8]. Human perception research suggests that delays under ~50 ms appear instantaneous to viewers [9].

C. Signal Processing Methods

Several DSP techniques enable meaningful audio-to-light mapping:

- **Amplitude detection**, mapping loudness to LED brightness [10]
- **Spectral band analysis**, mapping bass/mid/treble energy to color zones [11]
- **FFT-based spectral processing**, enabling fine-grained visualization [2]
- **Beat/onset detection**, creating rhythm-synchronized lighting [12]

The Raspberry Pi 5's processing capabilities allow these to be executed in real time.

D. Communication Between Raspberry Pi and Arduino

Most hybrid systems communicate over USB serial due to its reliability and minimal setup requirements. The Pi transmits color and animation data to the Arduino in compact packets, which the Arduino parses and renders [13], [14]. This approach minimizes latency while maintaining stable LED refresh rates.

E. LED Control and Lighting Technologies

WS2812B and APA102 LEDs are widely used for audio-reactive systems. APA102 LEDs support separate data and clock lines, enabling higher refresh rates, whereas WS2812B LEDs use a strict single-wire protocol [16], [17]. Proper electrical design is essential to ensure power stability and prevent signal degradation [18].

F. Prior Implementations and Design Challenges

Academic and hobbyist projects have validated the feasibility of Pi-Arduino hybrid lighting systems [6], [19]. Common challenges include latency reduction, electrical noise management, computational efficiency, and stable power distribution for large LED arrays. Evaluation typically involves measuring latency, CPU utilization, spectral accuracy, and subjective synchronization quality [20].

III. System Description

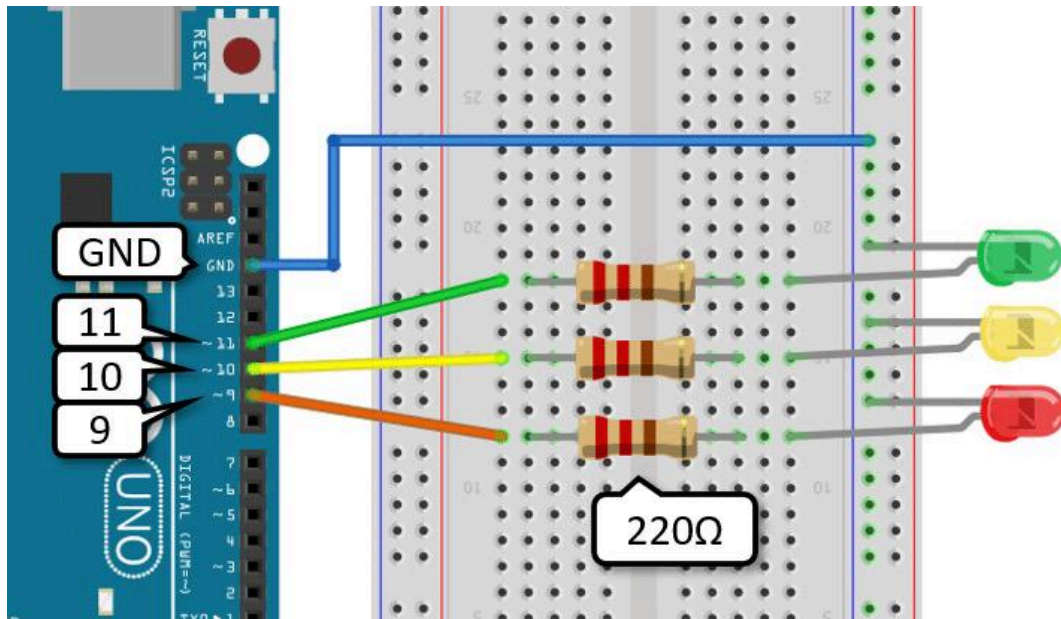
A. System Overview

Coloratura consists of two main subsystems:

1. **Input Subsystem**

- USB microphone connected to the Raspberry Pi 5
- Real-time DSP performed using Python (NumPy)

2. **Output Subsystem**



- Addressable RGB LED matrix driven by an Arduino
- Lighting mapped to amplitude, spectral bands, and rhythmic cues

B. Development Methodology

The project will be completed in sequential phases:

1. **Audio Configuration** – Acquire and test real-time audio input.
2. **DSP Implementation** – Implement FFT, amplitude tracking, and beat detection.
3. **Communication Pipeline** – Build a low-latency serial protocol for Pi-to-Arduino data transfer.
4. **LED Firmware Development** – Create Arduino firmware for rendering animations.
5. **Color-Mapping Design** – Develop mapping strategies for feature-to-color relationships.
6. **Latency Optimization** – Tune buffers, data formats, and LED refresh rates.
7. **Testing and Iteration** – Validate responsiveness, stability, and noise robustness.
8. **Documentation and Demonstration** – Produce full documentation and a live demo.

C. Expected Deliverables

- Functional real-time audio-reactive LED system
- Raspberry Pi DSP software
- Arduino LED driver firmware
- Full documentation and architecture diagrams
- Live demonstration of synchronized audiovisual behavior

D. Hardware and Software Requirements

Hardware: Raspberry Pi 5, Arduino Due, USB microphone, addressable LED matrix, 5V power supply, wiring, optional level shifters.

Software: Raspberry Pi OS, Python with NumPy and PySerial, Arduino IDE with FastLED or NeoPixel libraries.

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